

Fundamentals of Information Security

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Symmetric Cryptosystems

- Formal definition

Definition 1.1: A *cryptosystem* is a five-tuple $(\mathcal{P}, \mathcal{C}, \mathcal{K}, \mathcal{E}, \mathcal{D})$, where the following conditions are satisfied:

1. $\mathcal{P}$ is a finite set of possible *plaintexts*;
2. $\mathcal{C}$ is a finite set of possible *ciphertexts*;
3. $\mathcal{K}$, the *keyspace*, is a finite set of possible *keys*;
4. For each $K \in \mathcal{K}$, there is an *encryption rule* $e_K \in \mathcal{E}$ and a corresponding *decryption rule* $d_K \in \mathcal{D}$. Each $e_K : \mathcal{P} \rightarrow \mathcal{C}$ and $d_K : \mathcal{C} \rightarrow \mathcal{P}$ are functions such that $d_K(e_K(x)) = x$ for every plaintext element $x \in \mathcal{P}$.

Facts about $E_K()$

- It is *injective* (one-to-one)
 - i.e. $E_K(x_1) = E_K(x_2)$ iff $x_1 = x_2$
- Why?
 - If not, then Bob does not know if the ciphertext came from x_1 or x_2

Some traditional Cryptosystems

Threat Model

Assumptions:

- Attacker has full knowledge of the encryption/decryption algorithm
 - Attacker doesn't know the secret key
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- **Aim of the attacker:**
 - Recover the secret key, or,
 - Given the ciphertext, recover the original plaintext even if the key is unknown



Shift Cipher

- Let $\mathcal{P} = \mathcal{C} = \mathcal{K} = \{\text{Set of English Alphabets}\}$
- i.e., $|\mathcal{P}| = |\mathcal{C}| = |\mathcal{K}| = 26$
- **Algorithm** - Shift each letter in the plaintext by a position specified by the key to form the ciphertext
- Let $k = 5$
 - i.e., shift each letter in the plaintext by 5 letters to the right
- E.g., Plaintext = I AM BOND
Ciphertext = N FR GTSI

Shift Cipher Cryptanalysis

- Let us suppose

Plaintext – i am bond

Ciphertext - l dp erqg

- Can you guess the key ?
- Key = 3
- You just broke Julius Caesar's cipher 😊

How did we cryptanalyze ?

- Try “shift of 1” to check if a valid plaintext is produced
- Else, try shift of 2, 3 etc.
- Only 26 key choices to test and try, breaking is super easy !!
- What we did here was to **exhaustively search the keyspace**
 - Attack is known as **BRUTE FORCE ATTACK**

